PRAJJAWAL SINGH

singhprajjawal87@gmail.com | +91 8303159391 | LinkedIn | GitHub | LeetCode | CodeChef | Codeforces

SUMMARY

Backend Engineer with 1+ year of experience building scalable, production-grade backend systems and APIs with FastAPI, Django and Java/Spring Boot, alongside deploying cloud-hosted services and real-time data pipelines. Hands-on with REST API design, third-party integrations (webhooks, idempotency, retries), PostgreSQL/MySQL, Redis, Kafka, ClickHouse, AWS, Docker, Kubernetes (Helm, Ingress) and Nginx. Strong in DSA and CS fundamentals (1,000+ problems solved); LeetCode Guardian (rating 2130, top 1.92%). Daily user of AI coding tools (Claude Code, Cursor, Copilot) to ship quality code faster.

TECHNICAL SKILLS

Languages: Java (17+), Python, C++, SQL, JavaScript

Java & Frameworks: Spring Boot, Spring MVC, Dependency Injection, Hibernate/JPA, MapStruct, Liquibase/Flyway, JUnit

Backend & APIs: REST API Design, Django, FastAPI, GraphQL, OAuth2, JWT, Authentication & Authorization, Microservices

Third-Party Integrations: Webhooks / Callback Processing, Idempotency Keys, Retries, Reconciliation, Replay Handling, HMAC Signing, JWS/JWE, Payment Gateway & Partner API Integration

Databases & Caching: PostgreSQL, MySQL, Indexing & Query Optimisation (N+1), ClickHouse, ScyllaDB, MongoDB, Redis, Memcached

Messaging & Architecture: Apache Kafka, SQS, RabbitMQ, Event-Driven Architecture, Eventual Consistency, CQRS, Event Sourcing, Distributed Systems, Stream Processing, DB Replication, Scalability, Fault Tolerance

Cloud & DevOps: AWS (EKS, RDS, S3, Secrets Manager), CDN, Load Balancer, Docker, Kubernetes, Helm, Ingress, Nginx (reverse proxy), CI/CD, GitHub Actions

Observability: OpenTelemetry, SigNoz, Grafana, Structured Logging, Distributed Tracing, Metrics

OS / Tools: Linux, Shell, Git, GitHub, Claude Code, Cursor, GitHub Copilot

CS Fundamentals: DSA, OOP, Concurrency, HTTP, JSON, System Design (URL Shortener, Rate Limiter)

Design Patterns: Repository, Factory, Strategy, Observer, Singleton, MVC, CQRS, State Machine

EXPERIENCE

Backend Engineer - TailNode (Full-time) *Delhi, India | Jul 2025 - Present*

- Investigated slow shape loading and rising cloud costs across content delivery, coordinate payloads, S3 usage and Kafka replication traffic across Availability Zones.
- Implemented CDN caching: average load time fell from 5s to 2s (60%); monthly origin data-transfer costs fell from $500 to $300 (40%). Integrated a WAF to filter malicious requests and protect application endpoints.
- Removed redundant coordinates and optimised precision while preserving required shape accuracy; average payload fell from 2 MB to 800 KB (60%), improving shape download, rendering and interaction.
- Optimised stored data and applied appropriate S3 lifecycle policies, reducing monthly storage costs from $300 to $210 (30%).
- Moved a suitable Kafka workload to WarpStream, reducing cross-AZ replication traffic and monthly messaging costs from $700 to $420 (40%). Total spend across these categories fell from $1,500 to $930, saving $570/month (38%).
- Containerized services with Docker and deployed on Kubernetes using Helm, Ingress and load balancers, supporting a 99.9% uptime SLA; configured Nginx for high-volume, real-time data streams.
- Automated Kafka-to-ClickHouse ETL for 500K+ daily events, cutting reporting latency from hours to under 2 minutes; added replication, CQRS and Event Sourcing for high availability and audit-ready data.
- Developed REST and FastAPI endpoints (sub-50ms p99) secured via OAuth2/JWT, using Repository, Factory and Strategy patterns and dependency-injection-style layering.
- Led design reviews on PostgreSQL sharding/indexing, Redis caching, ScyllaDB writes and CAP trade-offs. Debugged production issues across logs, metrics and databases; used Claude Code, Cursor and Copilot with critical review.

PROJECTS

Social Media Application - Skyline | JavaScript, Node.js, React, MongoDB | GitHub

- Built a full-stack social platform with auth, posts and feeds; REST APIs integrated with a responsive React frontend, deployed with live hosting.
- Cut average feed query time by 70% with MongoDB compound indexes; implemented CI/CD via GitHub Actions.
- Applied MVC and Observer design patterns for a clean, scalable codebase serving 1,000+ concurrent users.

Quiz Competition Platform | JavaScript, Redis, Django, React | GitHub

- Delivered real-time leaderboard updates via Redis pub/sub (<100ms latency) for 500+ concurrent participants.
- Reduced page load by 45% through lazy loading, code splitting and Django query optimisation.
- Used the State Machine pattern for quiz session lifecycle management, ensuring data integrity.

EDUCATION

National Institute of Technology, Tiruchirappalli | B.Tech, Electronics & Communication Engineering *2021 - 2025*

CGPA: 7.91 / 10 | Don Bosco School | Class XII (PCM) - 91.8%

ACHIEVEMENTS

- LeetCode Guardian: rating 2130, top 1.92% globally; 953 problems solved (Easy 234 / Medium 490 / Hard 229); 330+ day streak, 198 active days and max streak of 41 in the past year.
- CodeChef 4-star, peak rating 1922; Codeforces Specialist, max rating ~1550; 1,000+ DSA problems solved across competitive platforms.